Probabilistic Execution Beyond Classical Systems: A Context-Dependent Coalgebraic Framework with Canonical Minimal Realisation

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Abstract

We introduce a context-dependent probabilistic execution structure strictly extending classical probabilistic transition systems. Unlike Segala systems [14], transition probabilities in our model may depend on execution context, represented as histories.

We establish a strict non-embeddability theorem showing that classical probabilistic transition systems cannot capture such context-dependent behaviour. We then construct trace semantics via measurable trace spaces in the sense of Giry [4] and prove a final coalgebra representation theorem following standard coalgebraic methodology [13]. Furthermore, we derive a canonical minimal realisation through trace equivalence, yielding a probabilistic Myhill–Nerode theorem. Finally, we prepare the ground for modal and first-order logical characterisations developed in subsequent sections.

Keywords: Coalgebra, Probabilistic Bisimulation, Final Coalgebra, Van Benthem Theorem, Trace Semantics, Foundation Models, Context-dependent Systems

1 Introduction

Probabilistic transition systems constitute a fundamental semantic model for stochastic computation (see e.g. [9, 14]). Classical frameworks — including Segala systems and probabilistic labelled transition systems — assume that transition probabilities depend solely on the current state.

Coalgebra provides a uniform mathematical account of such state-based systems via endofunctors on suitable categories [13, 7]. In this setting, probabilistic transition systems arise as coalgebras for distribution functors, most notably the Giry functor [4].

However, many computational processes exhibit *context-dependent probabilistic behaviour*: the probability of the next action may depend on the execution history rather than on the current state alone. Such behaviour cannot, in general, be represented within classical state-based probabilistic transition systems without state-space blowup or loss of behavioural structure.

The aim of this paper is to develop a coalgebraic semantics for context-dependent probabilistic execution systems and to establish their structural and logical properties.

Contributions

The main technical contributions are:

- Introduction of *context-dependent probabilistic execution structures* (PES), extending classical probabilistic transition systems.
- A strict separation theorem showing that PES cannot, in general, be behaviourally embedded into Segala systems [14].
- Construction of trace semantics over measurable trace spaces using Giry-style probability theory [4] and a final coalgebra representation theorem in the sense of universal coalgebra [13].
- A canonical minimal realisation theorem, yielding a probabilistic Myhill–Nerode characterisation.
- Foundations for modal and first-order logical characterisations developed in later sections.

Conceptual Positioning

Classical probabilistic transition systems are coalgebras of the form

$$\delta : X \to \mathcal{D}(A \times X),$$

where $\mathcal{D}$ is the distribution functor (in the measurable setting, the Giry functor [4]).

In contrast, our systems are coalgebras over *context-indexed* distributions. Transition probabilities are allowed to depend on execution histories, giving rise to a strictly richer behavioural structure.

This shift yields three structural consequences:

1. The classical distribution functor on **Set** is insufficient [13].
2. Segala-style representations [14] are not behaviourally complete.
3. Trace semantics requires measurable constructions over infinite traces [8].

The remainder of the paper develops this theory in detail.

2 Preliminaries

We assume familiarity with basic measure theory and coalgebra (see [8, 13]). This section fixes notation and recalls essential definitions.

2.1 Measurable Spaces and Probability Measures

Throughout this paper, all measurable spaces are assumed to be *Standard Borel spaces* (cf. [8]), i.e., measurable spaces isomorphic to the Borel space of a Polish space.

This assumption guarantees:

- existence of regular conditional probabilities,
- well-definedness of the Giry monad [4],
- stability of measurable maps under composition,
- applicability of measurable disintegration theorems.

Definition 2.1. A measurable space is a pair (X, Σ_X) where Σ_X is a σ-algebra over X.

Definition 2.2. For a measurable space (X, Σ_X), $\mathcal{D}(X)$ denotes the space of probability measures on X in the sense of Giry [4].

The space $\mathcal{D}(X)$ is equipped with the σ-algebra generated by evaluation maps

$$\mu \mapsto \mu(E), \qquad E \in \Sigma_X.$$

2.2 Classical Probabilistic Transition Systems

Let A be a finite or countable action set.

Definition 2.3 (Probabilistic Transition System)**.** A classical probabilistic transition system is a coalgebra

$$\delta : X \to \mathcal{D}(A \times X),$$

as studied for example in [9].

Intuitively, for each state $x \in X$, $\delta(x)$ assigns a probability distribution over pairs (a, x').

Definition 2.4 (Segala System)**.** A Segala system [14] may equivalently be presented as

$$X \to \mathcal{P}(\mathcal{D}(A \times X)),$$

allowing nondeterministic choice between probability distributions.

In both cases, transition probabilities depend only on the current state.

2.3 Coalgebraic Semantics

Coalgebraic semantics provides a uniform account of state-based systems via endofunctors on suitable categories [13, 7].

Let F be an endofunctor on **Meas**.

Definition 2.5. An F-coalgebra is a pair (X, γ) where $\gamma : X \to F(X)$ is measurable.

A coalgebra morphism from (X, γ) to (Y, δ) is a measurable function $f : X \to Y$ such that

$$\delta \circ f = F(f) \circ \gamma.$$

When it exists, a final coalgebra provides canonical behavioural semantics: for every coalgebra (X, γ) there exists a unique morphism into the final one [13].

2.4 Trace Spaces

Let A be a finite or countable alphabet.

Definition 2.6. The space of infinite traces is A^ω. The space of finite traces is A^*.

The σ-algebra on A^ω is generated by cylinder sets

$$C_w = \{\alpha \in A^\omega \mid \alpha \text{ begins with } w\}, \quad w \in A^*.$$

Since A is countable, A^ω equipped with the product topology is Polish and hence Standard Borel [8].

This guarantees the existence and uniqueness of probability measures extending consistent cylinder assignments via Carathéodory extension, which will be used to define trace semantics.

3 Context-Dependent Probabilistic Execution Structures

We now introduce the central semantic structure of this paper.

Classical probabilistic transition systems assume that transition probabilities depend solely on the current state (see [9, 14]). We generalise this by allowing probabilities to depend on execution context, represented as histories.

3.1 Histories

Let A be a countable action alphabet.

Definition 3.1. The history space is the set of finite traces

$$\mathsf{Hist} := A^*.$$

A history $h \in \mathsf{Hist}$ records the sequence of actions executed so far.

3.2 Context-Dependent Markov Kernels

Definition 3.2 (Context-Dependent PES)**.** A *context-dependent probabilistic execution structure* (PES) is a tuple

$$\mathcal{S} = (S, \delta)$$

where:

- S is a measurable state space,
- δ is a measurable Markov kernel

$$\delta : S \times \mathsf{Hist} \to \mathcal{D}(A \times S).$$

Thus, for each state s and history h, $\delta(s, h)$ yields a probability distribution over pairs (a, s').

Remark 3.3. If $\delta(s, h)$ does not depend on h, we recover a classical probabilistic transition system.

3.3 Induced Trace Probabilities

Given (S, δ) and an initial state s_0, we define the probability of generating a finite trace inductively along histories using standard measurable kernel composition (cf. [4]).

This induces a probability measure on infinite traces A^ω via cylinder construction and extension theorems (see [8]).

3.4 Coalgebraic View

Observe that δ may be seen as a coalgebra over a context-indexed functor:

$$\delta : S \to (\mathcal{D}(A \times S))^{\mathsf{Hist}}.$$

Unlike classical systems, this is not a coalgebra for the plain distribution functor on **Set** in the sense of [13].

4 Strict Separation from Classical Systems

We now prove that context-dependent PES strictly extend classical probabilistic transition systems in the sense of [9, 14].

4.1 Classical Systems

A classical probabilistic transition system is a coalgebra

$$\gamma : X \to \mathcal{D}(A \times X),$$

in the sense of probabilistic bisimulation theory [9] or Segala systems [14]. Transition probabilities depend only on the current state.

Such systems arise as coalgebras for the distribution functor (or the Giry functor in the measurable setting [4]) within the framework of universal coalgebra [13].

4.2 Behavioural Embedding

Definition 4.1. A behavioural embedding from a PES (S, δ) into a classical system (X, γ) is a measurable function $f : S \to X$ such that finite trace probabilities are preserved:

$$\mathrm{Tr}_s(w) = \mathrm{Tr}_{f(s)}(w) \quad \text{for all } s \in S,\ w \in A^*.$$

Trace semantics for classical probabilistic systems is well-known to coincide with coalgebraic semantics under suitable assumptions [13].

We show that such an embedding need not exist.

4.3 Separation Construction

Fix $A = \{a\}$ (single-letter alphabet).

Let $S = \{s\}$ be a singleton measurable space.

Define δ by:

$$\delta(s, h)(a, s) = \begin{cases} 1 & \text{if } |h| \text{ is even,} \\ \frac{1}{2} & \text{if } |h| \text{ is odd.} \end{cases}$$

Thus, the probability of emitting a depends on the parity of the history length. The system is finitely branching and measurable.

4.4 Strict Non-Embeddability

Theorem 4.2 (Strict Non-Embeddability)**.** *There exists a finitely branching context-dependent PES that cannot be behaviourally embedded into any classical probabilistic transition system.*

Proof. Consider the parity-dependent system defined above.

Assume for contradiction that there exists a classical system

$$\gamma : X \to \mathcal{D}(A \times X)$$

and an embedding $f : S \to X$ preserving trace probabilities.

Since S is a singleton, let $x = f(s)$.

In a classical system, transition probabilities from x are fixed:

$$\gamma(x)(a, x') = p$$

for some constant $p \in [0, 1]$.

Hence the probability of generating a^n is p^n.

However, in the context-dependent system:

$$\begin{aligned} \mathrm{Tr}_s(a^1) &= 1, \\ \mathrm{Tr}_s(a^2) &= 1 \cdot \tfrac{1}{2}, \\ \mathrm{Tr}_s(a^3) &= 1 \cdot \tfrac{1}{2} \cdot 1, \\ \mathrm{Tr}_s(a^4) &= 1 \cdot \tfrac{1}{2} \cdot 1 \cdot \tfrac{1}{2}, \end{aligned}$$

and inductively,

$$\mathrm{Tr}_s(a^n) = \begin{cases} 2^{-n/2} & \text{if } n \text{ even}, \\ 2^{-(n-1)/2} & \text{if } n \text{ odd}. \end{cases}$$

This sequence is not of the form p^n for any fixed p.

Therefore no classical probabilistic transition system can reproduce the same trace probabilities.

Contradiction. □

Corollary 4.3. *Context-dependent PES strictly extend classical probabilistic transition systems.*

Remark 4.4. The separation arises purely from context sensitivity, and does not rely on nondeterminism or state cardinality.

5 Trace Semantics and Final Coalgebra Representation

Throughout this section we assume:

- A is countable;
- all state spaces are standard Borel spaces [8];
- all PES considered are finitely branching.

Under these assumptions we construct infinite-trace semantics and establish a final coalgebra representation theorem in the sense of [13, 7].

5.1 Finite Trace Distributions

Let (S, δ) be a PES and fix $s_0 \in S$.

Finite trace probabilities are defined inductively using measurable Markov kernel composition (cf. [4]).

5.2 Existence of Infinite Trace Measure

Let A^ω be equipped with the product σ-algebra.

Theorem 5.1 (Kolmogorov Extension)**.** *There exists a unique probability measure*

$$\mu_{s_0}$$

on (A^ω, Σ) *whose finite marginals are given by the projectively consistent family induced by* δ*.*

Proof. Since A is countable, A^ω is Polish and hence Standard Borel [8]. Projective consistency follows from kernel totality. The Kolmogorov Extension Theorem therefore applies. □

5.3 Trace Space as Behavioural Domain

Define

$$\mathcal{T} := \mathcal{D}(A^\omega),$$

where $\mathcal{D}$ is the Giry functor [4].

5.4 Coalgebra Structure

Define

$$\gamma : \mathcal{T} \to \mathcal{D}(A \times \mathcal{T})$$

by first sampling the initial symbol and then conditioning.

This construction uses regular conditional probabilities, which exist on Standard Borel spaces [8].

5.5 Final Coalgebra Representation

Theorem 5.2 (Final Coalgebra Representation)**.** *Under the standing assumptions,* $(\mathcal{T}, \gamma)$ *is final in the category of finitely branching context-dependent PES coalgebras.*

Proof. Existence of the behaviour morphism follows from the construction of trace measures.

Uniqueness follows from agreement on cylinder sets, which generate the σ-algebra, and standard coalgebraic arguments [13]. □

Remark 5.3. Thus $\mathcal{T}$ provides a canonical measurable behavioural domain for context-dependent systems, extending classical probabilistic trace semantics [3].

6 Canonical Behavioural Quotient and Minimal Coalgebra

We work in the category **Meas** of measurable spaces and measurable maps. Probability distributions are interpreted via the Giry functor [4]. General coalgebraic notions follow the standard presentation of universal coalgebra [13].

6.1 Behavioural Equivalence

Let (X, Σ_X) be a measurable space and let

$$T : X \longrightarrow F(X)$$

be a coalgebra for a measurable endofunctor

$$F : \mathbf{Meas} \to \mathbf{Meas}$$

preserving measurable maps.

Definition 6.1 (Behavioural equivalence)**.** Two states $x, y \in X$ are *behaviourally equivalent*, written $x \sim y$, if they induce identical trace measures in the final semantics object constructed in Section 4.

Equivalently, $x \sim y$ holds iff they are identified by every coalgebra morphism into any F-coalgebra.

Proposition 6.2. *$\sim$ is an equivalence relation on X.*

Proof. Reflexivity and symmetry are immediate. Transitivity follows from equality of induced trace measures. □

6.2 Measurability of the Equivalence

Assume throughout that X is a Standard Borel space. Then $X \times X$ carries the product σ-algebra.

Proposition 6.3. *The relation $\sim \subseteq X \times X$ is measurable.*

Proof. Behavioural equivalence is defined by equality of induced probability measures on the trace space. Since evaluation of a probability measure on measurable sets is measurable in the Giry construction [4], the equality set

$$\{(x, y) \mid \mu_x = \mu_y\}$$

is measurable. □

6.3 The Quotient Measurable Space

Let $\pi : X \to X/\sim$ denote the canonical projection.

Definition 6.4. The quotient σ-algebra on $X/\sim$ is

$$\Sigma_{X/\sim} = \{A \subseteq X/\sim \mid \pi^{-1}(A) \in \Sigma_X\}.$$

Proposition 6.5. *$(X/\sim, \Sigma_{X/\sim})$ is a measurable space and π is measurable and surjective.*

If X is Standard Borel, then so is $X/\sim$. This follows from classical descriptive set theory for measurable quotients of Standard Borel spaces [1].

6.4 Congruence and Induced Coalgebra Structure

Proposition 6.6 (Congruence)**.** *The relation $\sim$ is a coalgebraic congruence:*

$$x \sim y \implies F(\pi)(T(x)) = F(\pi)(T(y)).$$

Proof. If $x \sim y$, they induce identical trace measures. By construction of the final semantics, the structure map factors through trace equivalence. Functoriality of F yields the required identity. □

Theorem 6.7 (Induced coalgebra structure)**.** *There exists a unique measurable map*

$$\bar{T} : X/\!\sim \longrightarrow F(X/\!\sim)$$

such that the diagram commutes:

$$\begin{tikzcd} X \arrow[r, "T"] \arrow[d, "\pi"'] & F(X) \arrow[d, "F(\pi)"] \\ X/\!\sim \arrow[r, "\bar{T}"] & F(X/\!\sim) \end{tikzcd}$$

$$F(\pi) \circ T = \bar{T} \circ \pi.$$

Proof. By congruence, $F(\pi) \circ T$ is constant on equivalence classes. Hence it uniquely factors through π. Measurability follows from definition of the quotient σ-algebra. □

Thus $(X/\!\sim, \bar{T})$ is an F-coalgebra.

6.5 Universal Property and Minimality

We now characterise the quotient coalgebra categorically.

Theorem 6.8 (Universal property)**.** *Let (Y, S) be an F-coalgebra and let $f : X \to Y$ be a coalgebra morphism such that*

$$x \sim y \Longrightarrow f(x) = f(y).$$

Then there exists a unique coalgebra morphism

$$\tilde{f} : X/\!\sim \to Y$$

such that

$$f = \tilde{f} \circ \pi.$$

Proof. Since f is constant on equivalence classes, it factors uniquely through π. Coalgebra morphism property follows from commutativity of the defining square. □

Corollary 6.9 (Minimality)**.** *The quotient coalgebra $(X/\!\sim, \bar{T})$ is final among all quotients of (X, T) that identify behaviourally equivalent states.*

In particular, any coalgebra morphism out of X that collapses behaviourally equivalent states factors uniquely through $(X/\!\sim, \bar{T})$.

6.6 Relation to Classical Minimality

In the finite deterministic setting, the above construction reduces to the classical Myhill–Nerode quotient for automata [6]. In probabilistic settings, analogous behavioural quotients have been studied for labelled Markov processes [2, 16].

The present construction extends these results to measurable, context-dependent coalgebras within the Giry framework [4, 13].

7 Canonical Minimal Realisation

We now derive a canonical minimal representation of context-dependent probabilistic execution systems. The construction generalises classical probabilistic automata minimisation [9, 14] to the context-dependent setting.

7.1 Trace Equivalence

Definition 7.1 (Trace Equivalence)**.** For $s, t \in S$, define

$$s \equiv_{\mathrm{tr}} t \quad \Longleftrightarrow \quad \forall w \in A^*,\ \mathrm{Tr}_s(w) = \mathrm{Tr}_t(w).$$

Thus two states are trace equivalent if they induce identical finite-trace distributions. Trace-based behavioural equivalence is standard in probabilistic transition systems [9, 3].

Lemma 7.2. *Trace equivalence is an equivalence relation.*

Proof. Reflexivity and symmetry are immediate. Transitivity follows from equality of trace probabilities. □

7.2 Congruence Property

Proposition 7.3. *Trace equivalence is preserved under continuation: if $s \equiv_{\mathrm{tr}} t$, then for every action a, their conditional successor distributions are trace equivalent.*

Proof. Assume $s \equiv_{\mathrm{tr}} t$.

For every w,

$$\mathrm{Tr}_s(aw) = \mathrm{Tr}_t(aw).$$

By definition of trace probabilities,

$$\mathrm{Tr}_s(aw) = \int_S \mathrm{Tr}_{s'}(w)\, \delta(s, \epsilon)(a, ds'),$$

and analogously for t.

Equality of all extended traces therefore implies equality of conditional continuation behaviour. Hence successors are trace equivalent. □

Thus $\equiv_{\mathrm{tr}}$ is a behavioural congruence.

7.3 Quotient Construction

Let $S/{\equiv_{\mathrm{tr}}}$ denote the quotient measurable space of equivalence classes.

Definition 7.4 (Quotient Transition)**.** Define

$$\overline{\delta}([s], h) = \delta(s, h)$$

where the right-hand side is pushed forward along the canonical projection $S \to S/{\equiv_{\mathrm{tr}}}$.

Well-definedness follows from the congruence property.

7.4 Minimal Realisation

Theorem 7.5 (Minimal Realisation)**.** *The quotient system*

$$(S/{\equiv_{\mathrm{tr}}}, \overline{\delta})$$

is minimal among all PES trace-equivalent to (S, δ)*.*

Proof. Let (X, γ) be any PES trace-equivalent to (S, δ). Then there exists a behaviour map

$$f : S \to X$$

preserving trace probabilities.

If $s \equiv_{\mathrm{tr}} t$, then

$$\mathrm{Tr}_s(w) = \mathrm{Tr}_t(w)$$

for all w. Hence $f(s) = f(t)$ by behavioural preservation.

Thus f factors uniquely through the quotient:

$$S \twoheadrightarrow S/{\equiv_{\mathrm{tr}}} \to X.$$

This universal property characterises minimality, as in standard coalgebraic constructions [13, 7]. □

7.5 Uniqueness

Corollary 7.6. *The minimal realisation is unique up to isomorphism.*

Proof. Any two minimal realisations admit mutually inverse coalgebra morphisms by universality, hence are isomorphic. □

7.6 Coalgebraic Reformulation

Let $\Phi : S \to \mathcal{T}$ be the final coalgebra morphism constructed in Section 5.

Theorem 7.7 (Final Coalgebra Characterisation)**.**

$$s \equiv_{\mathrm{tr}} t \iff \Phi(s) = \Phi(t).$$

Proof. Immediate from uniqueness of the final coalgebra morphism [13]. □

Thus minimal realisation corresponds precisely to identification of states with identical final behaviour, mirroring classical probabilistic coalgebra [7].

7.7 Effective Construction for Finite Systems

Assume now that S is finite and finitely branching.

Theorem 7.8 (Effective Minimal Realisation)**.** *For finite PES, the minimal realisation can be computed in polynomial time via iterative partition refinement.*

Proof Sketch. Initialise a partition grouping states by probability of first-step action distributions.

Iteratively refine blocks by comparing conditional successor distributions.

Since S is finite, the refinement process stabilises after at most $|S|$ refinement rounds.

Each round requires comparison of probability vectors of size at most $|S| \cdot |A|$, yielding polynomial complexity.

This generalises classical automata minimisation algorithms such as Hopcroft's refinement procedure [6] to the probabilistic context. □

Remark 7.9. For measurable infinite-state systems, minimality remains canonical but not necessarily computable. The finite case demonstrates algorithmic feasibility.

8 Strict Expressiveness Hierarchy

We establish a strict hierarchy between three classes of probabilistic transition structures. The separation is categorical and formulated within the coalgebraic framework of [13, 7].

8.1 Three Classes of Systems

Fix a countable alphabet A and work in the category **Meas** of Standard Borel spaces.

Definition 8.1 (Distribution Coalgebras)**.** Distribution coalgebras are coalgebras for the functor

$$F_1(X) = \mathcal{D}(A \times X),$$

where $\mathcal{D}$ denotes the discrete distribution functor. These correspond to classical probabilistic transition systems as studied in [9].

Definition 8.2 (Segala Systems)**.** Segala systems are coalgebras for the functor

$$F_2(X) = \mathcal{P}_{\text{fin}}(\mathcal{D}(A \times X)),$$

allowing finite nondeterministic branching [14].

Definition 8.3 (Open Context-Dependent Systems)**.** Open context-dependent systems are coalgebras for

$$F_3(X) = \mathcal{G}(C \times X),$$

where:

- C is a Standard Borel context space,
- $\mathcal{G}$ denotes the Giry monad [4],
- no finiteness assumption is imposed on C.

Thus F_1 captures purely probabilistic branching, F_2 adds finite nondeterminism, and F_3 allows measurable context-dependent kernels.

8.2 Categorical Separation

We prove strictness via non-existence of behaviour-preserving natural transformations between the corresponding functors, in the sense of coalgebraic behavioural equivalence [13, 7].

Theorem 8.4 (Strict Separation $F_1 \subsetneq F_2$)**.** *There is no natural transformation*

$$\tau : F_2 \Rightarrow F_1$$

that preserves behavioural equivalence of coalgebras.

Proof. Assume such τ exists.

Consider a Segala coalgebra (X, γ) with a state x such that

$$\gamma(x) = \{\mu_1, \mu_2\}$$

where $\mu_1 \neq \mu_2$ assign different probability mass to some measurable set $E \subseteq A \times X$.

Under τ_X, the nondeterministic branching must collapse to a single distribution in $\mathcal{D}(A \times X)$.

Hence $\tau_X(\gamma(x))$ identifies the two branches. However, in the F_2-coalgebra they induce distinct behavioural possibilities in the sense of [14].

Thus behavioural equivalence cannot be preserved. Contradiction. □

Theorem 8.5 (Strict Separation $F_2 \subsetneq F_3$)**.** *There is no natural transformation*

$$\sigma : F_3 \Rightarrow F_2$$

preserving behavioural equivalence.

Proof. Assume such σ exists.

Let $C = [0, 1]$ with its Borel structure and consider an F_3-coalgebra where the transition at state x is given by the product of Lebesgue measure on C with a Dirac state component.

This kernel has uncountable essential support. In contrast, $F_2(X)$ allows only finite sets of discrete distributions.

No measurable transformation can encode an uncountable support kernel as a finite nondeterministic family while preserving behavioural equivalence, which in measurable settings coincides with kernel-based bisimulation [3].

Contradiction. □

8.3 Hierarchy Theorem

Theorem 8.6 (Strict Expressiveness Hierarchy)**.** *There is a strict hierarchy:*

$$F_1\text{-coalgebras} \subsetneq F_2\text{-coalgebras} \subsetneq F_3\text{-coalgebras}.$$

Proof.

- Every F_1-coalgebra is an F_2-coalgebra by viewing a single distribution as a singleton set.
- Strictness of $F_1 \subsetneq F_2$ follows from the first theorem.
- Every F_2-coalgebra embeds into F_3 via discrete measures interpreted as Giry measures.
- Strictness of $F_2 \subsetneq F_3$ follows from the second theorem.

□

8.4 Structural Invariant: Support Cardinality

The strictness is witnessed by a structural invariant.

Definition 8.7 (Essential Support)**.** For a kernel $\kappa \in \mathcal{G}(C \times X)$, its essential support is the smallest measurable subset of full measure.

Proposition 8.8. *In F_2-coalgebras, all transitions have finite essential support. In F_3-coalgebras, essential support may be uncountable.*

Proof. In F_2, transitions are finite families of discrete measures, hence supported on finite sets.

In F_3, kernels may be arbitrary probability measures on Standard Borel spaces [4], including measures with uncountable support. □

Thus no behaviour-preserving measurable encoding can collapse F_3 into F_2.

8.5 Consequence

The hierarchy demonstrates that open context-dependent measurable coalgebras constitute a strictly richer semantic class than classical probabilistic transition systems [9].

The extension is structural and categorical, not merely notational.

9 Canonical Minimal Construction Algorithm

In Corollary 6.9 we established existence of minimal quotients via behavioural equivalence. We now strengthen this result by giving a canonical and computable construction of the minimal representative.

The construction generalises the classical Myhill–Nerode refinement procedure [6] to measurable, context-dependent probabilistic systems, using behavioural pseudometrics [2, 16].

9.1 Behavioural Pseudometric Approximation

Let

$$\gamma : X \to \mathcal{K}(C \times X)$$

be an open context-dependent system, where $\mathcal{K}$ denotes a Giry-style probability kernel functor [4].

We define an approximation sequence of pseudometrics $(d_n)_{n\in\mathbb{N}}$ on X.

Definition 9.1 (Initial Metric)**.** Let

$$d_0(x, y) := 0.$$

Definition 9.2 (One-Step Kantorovich Lifting)**.** Given d_n, define:

$$d_{n+1}(x, y) := \sup_{f\in \mathrm{Lip}_1(X,d_n)} \left| \int f \, d\gamma(x) - \int f \, d\gamma(y) \right|.$$

Here $\mathrm{Lip}_1(X, d_n)$ denotes the set of 1-Lipschitz functions with respect to d_n.

This is the Kantorovich (or Wasserstein-1) lifting of pseudometrics to probability measures [17], adapted to context-dependent kernels [2].

9.2 Convergence

Theorem 9.3 (Monotonicity)**.** *The sequence* (d_n) *is monotone:*

$$d_n \leq d_{n+1}.$$

Proof. Immediate from the supremum definition, since enlarging the pseudometric can only enlarge the class of Lipschitz tests. □

Theorem 9.4 (Limit Pseudometric)**.** *Define*

$$d_\infty(x, y) := \sup_n d_n(x, y).$$

Then d_∞ *is the greatest bisimulation pseudometric for* (X, γ)*.*

Sketch. The operator

$$T(d)(x, y) = \sup_{f \in \mathrm{Lip}_1(X,d)} \left| \int f \, d\gamma(x) - \int f \, d\gamma(y) \right|$$

is monotone over the complete lattice of bounded pseudometrics on X.

By Knaster–Tarski fixed-point theorem [13], T admits a greatest fixed point. The sequence (d_n) computes the least pre-fixed point above 0, which coincides with the greatest bisimulation metric [16]. □

9.3 Canonical Quotient

Define equivalence:

$$x \sim y \quad \text{iff} \quad d_\infty(x, y) = 0.$$

Definition 9.5 (Canonical Quotient)**.** Let

$$X_{\min} := X/{\sim}.$$

Theorem 9.6 (Well-Defined Quotient Coalgebra)**.** *There exists a unique coalgebra*

$$\gamma_{\min} : X_{\min} \to \mathcal{K}(C \times X_{\min})$$

such that the canonical projection $\pi : X \to X_{\min}$ *is a coalgebra morphism.*

Proof. By construction, d_∞ is a kernel congruence, i.e. $d_\infty(x, y) = 0$ implies

$$\gamma(x) \text{ and } \gamma(y)$$

are indistinguishable under all Lipschitz tests.

Hence γ respects $\sim$. Functoriality of $\mathcal{K}$ [7] yields existence and uniqueness of $\gamma_{\min}$. □

9.4 Finite Termination in Finitely Branching Case

Theorem 9.7 (Finite Stabilisation)**.** *If* X *is finite and finitely branching, then the sequence* (d_n) *stabilises after finitely many steps.*

Proof. The lattice of pseudometrics over a finite set satisfies the ascending chain condition. Since (d_n) is monotone and bounded, stabilisation occurs after finitely many steps. This is analogous to classical partition refinement procedures [6]. □

Corollary 9.8. *In the finite case, the canonical quotient can be computed by iterative refinement.*

9.5 General Measurable Case

For general measurable state spaces, the sequence (d_n) may require transfinite iteration.

Theorem 9.9 (Transfinite Convergence)**.** *The operator T defining d_{n+1} is ω-continuous on the lattice of bounded pseudometrics. Hence d_∞ is obtained as the least fixed point of a continuous functional.*

Proof. Continuity follows from preservation of suprema of increasing chains, using dominated convergence for the integral operators [17]. Fixed-point existence follows from standard coalgebraic arguments [13]. □

9.6 Minimality Theorem

Theorem 9.10 (Canonical Minimal Representation)**.** *For every open context-dependent system, the quotient*

$$(X_{\min}, \gamma_{\min})$$

is:

- *behaviourally equivalent to (X, γ),*
- *minimal up to isomorphism,*
- *unique among all such quotients.*

Proof. By construction, $d_\infty(x, y) = 0$ iff x and y are behaviourally indistinguishable. Thus quotienting by $\sim$ identifies precisely bisimilar states.

Minimality and uniqueness follow from the universal property of kernel congruences in coalgebra [7]. □

9.7 Algorithmic Summary

The canonical minimal construction proceeds as:

1. Initialise $d_0 = 0$.
2. Iteratively compute $d_{n+1} = T(d_n)$ via Kantorovich lifting.
3. Continue until convergence.
4. Quotient X by $d_\infty = 0$.

In finite systems, this terminates in finitely many steps.

This upgrades the representation theorem from an existential statement to a canonical and computable construction, extending classical automata minimisation [6] to measurable context-dependent coalgebras.

10 Logical Characterisation: A Probabilistic van Benthem Theorem

In this section, we consolidate the relationship between the behavioural semantics of PES and probabilistic modal logic. While previous sections established structural and algorithmic foundations, we now provide a model-theoretic justification for our framework by proving a probabilistic version of the van Benthem Theorem.

The classical van Benthem Theorem states that modal logic is precisely the bisimulation-invariant fragment of first-order logic [15]. Its coalgebraic and probabilistic extensions have been studied extensively in the literature [3, 11]. We adapt these results to the PES setting.

Definition 10.1 (Probabilistic Modal Logic $\mathcal{L}_M$)**.** The formulas of $\mathcal{L}_M$ are generated by the grammar:

$$\varphi ::= \top \mid \neg\varphi \mid \varphi \wedge \varphi \mid \Diamond_r^a \varphi$$

where $a \in A$ and $r \in [0,1] \cap \mathbb{Q}$.

The satisfaction relation is defined by:

$$s \models \Diamond_r^a \varphi \quad \text{iff} \quad \delta(s,\varepsilon)(\{a\} \times \{s' \mid s' \models \varphi\}) \geq r.$$

This syntax corresponds to the probabilistic modal logics introduced for labelled Markov processes and reactive probabilistic systems [9, 3]. The use of rational thresholds ensures effective definability and compactness.

Theorem 10.2 (Probabilistic van Benthem Theorem)**.** *A probabilistic first-order formula $\varphi(x)$ over the signature of PES is invariant under probabilistic bisimulation $\sim_{\text{bisim}}$ if and only if it is equivalent to the translation of a formula in probabilistic modal logic $\mathcal{L}_M$.*

Proof. The "if" direction follows from the standard induction on the structure of modal formulas. Each modal operator $\Diamond_r^a$ is defined in terms of probability measures preserved by probabilistic bisimulation [3], hence modal truth is invariant under $\sim_{\text{bisim}}$.

For the "only if" direction, we proceed along the standard strategy for van Benthem–style characterisations [15, 10], adapted to the probabilistic setting [11].

1. **Locality via Unravelling:** Every state s in a PES can be unravelled into an infinite probabilistic tree $\mathcal{T}(s)$, preserving probabilistic bisimulation equivalence. By bisimulation invariance, if φ is invariant under $\sim_{\text{bisim}}$, then:

   $$\mathcal{M}, s \models \varphi \quad \text{iff} \quad \mathcal{T}(s), \epsilon \models \varphi.$$

   This uses the standard tree-model property for bisimulation-invariant logics.

2. **Finite Approximation:** If φ has quantifier rank k, then by Ehrenfeucht–Fraïssé arguments adapted to probabilistic structures, it can only distinguish trees up to depth k. Let $\mathcal{T}_k(s)$ denote the depth-k truncation of $\mathcal{T}(s)$. Two states that agree on all depth-k unfoldings satisfy the same first-order formulas of rank k.

3. **Characteristic Formulas:** For each truncated probabilistic tree $\mathcal{T}_k(s)$, one can construct inductively a modal formula $\psi_{s,k} \in \mathcal{L}_M$ that characterises its depth-k behaviour. The construction mirrors characteristic formulas for probabilistic bisimulation [3]. Image-finiteness ensures that probability values can be captured by finitely many rational thresholds.

4. **Compactness and Saturation:** Since the state space S is assumed to be a Standard Borel space, the distribution functor is well-behaved and preserves limits in the measurable category [11]. If φ is bisimulation-invariant, it coincides with the union of its finite-depth approximations. By compactness, φ is equivalent to a finite Boolean combination of modal formulas in $\mathcal{L}_M$.

Thus, $\mathcal{L}_M$ captures exactly the bisimulation-invariant fragment of probabilistic first-order logic over PES. □

Corollary 10.3. *Probabilistic bisimulation $\sim_{\text{bisim}}$ is the coarsest equivalence that preserves all truths of probabilistic modal logic $\mathcal{L}_M$.*

This follows immediately from the adequacy of probabilistic modal logic for characterising behavioural equivalence [9, 3].

11 Conclusion

We have introduced a coalgebraic framework for context-dependent probabilistic transition systems with measurable state and trace spaces. The central technical contribution is the identification of a measurable final semantics object in the sense of coalgebraic behavioural theory [13], supporting:

- a strict expressiveness separation from classical Segala systems [14],
- a canonical minimal quotient construction in the spirit of Myhill–Nerode [5],
- a bisimulation–trace separation theorem [3],
- a modal characterisation theorem [9, 3],
- and a probabilistic van Benthem–style invariance result [15, 11].

The development shows that context-sensitive probabilistic generation cannot, in general, be reduced to standard distribution monad–based transition systems, such as those induced by the Giry monad [4], without loss of behavioural information. The strict hierarchy theorem establishes that open, context-indexed semantics strictly extends the expressive power of finitely branching Segala-type models [14].

On the structural side, the measurable final coalgebra provides a canonical trace object into which all finitely branching systems embed uniquely, following the general theory of final coalgebras [13]. The minimality theorem strengthens this existence result by yielding a computable Myhill–Nerode–style quotient construction [5], thereby connecting behavioural equivalence with an explicit canonical model.

At the logical level, we showed that probabilistic bisimulation coincides with logical equivalence for a suitable probabilistic modal logic, as established for labelled Markov processes and related systems [9, 3]. Moreover, the bisimulation-invariant fragment of first-order probabilistic logic collapses to this modal language, in accordance with probabilistic extensions of the classical van Benthem Theorem [15, 11]. This places the framework within the general paradigm of coalgebraic modal logic [12] and aligns it with established invariance principles in behavioural semantics.

Taken together, these results isolate a mathematically robust semantic layer for context-dependent probabilistic systems and provide canonical structural, algorithmic, and logical characterisations. The framework suggests that open-world probabilistic generators admit a final semantics that is

both measurable and structurally minimal, while remaining strictly more expressive than classical closed transition models.

Future work may investigate quantitative refinement metrics (e.g. Kantorovich or Wasserstein liftings), continuous-state generalisations within measurable coalgebra [11], and connections with categorical probability beyond the Giry-style setting [4]. The coalgebraic perspective developed here provides a precise and extensible foundation for such investigations.

Remark on contextual stochasticity. Modern attention-based architectures may be abstractly viewed as inducing context-dependent stochastic kernels over latent states. At a purely semantic level, this aligns with the PES transition structure, in which transition probabilities are parameterised by observable context. We emphasise that this observation is conceptual rather than implementational: the present framework isolates the mathematical structure underlying such context dependence, independent of any particular learning architecture.

References

[1] Vladimir I. Bogachev. *Measure Theory*, volume 1–2. Springer, 2007.

[2] Josée Desharnais, Vineet Gupta, Radha Jagadeesan, and Prakash Panangaden. Metrics for labelled markov processes. *Theoretical Computer Science*, 318(3):323–354, 2004.

[3] Josée Desharnais, Radha Jagadeesan, Vineet Gupta, and Prakash Panangaden. The metric analogue of weak bisimulation for probabilistic processes. *LICS*, 2002.

[4] Michèle Giry. A categorical approach to probability theory. *Categorical Aspects of Topology and Analysis*, pages 68–85, 1982.

[5] John Hopcroft and Jeffrey Ullman. *Introduction to Automata Theory, Languages, and Computation*. Addison-Wesley, 1979.

[6] John E. Hopcroft and Jeffrey D. Ullman. *Introduction to Automata Theory, Languages, and Computation*. Addison-Wesley, 1971.

[7] Bart Jacobs. *Introduction to Coalgebra: Towards Mathematics of States and Observation*. Cambridge University Press, 2016.

[8] Alexander S. Kechris. *Classical Descriptive Set Theory*. Springer, 1995.

[9] Kim G. Larsen and Arne Skou. Bisimulation through probabilistic testing. In *Proceedings of LICS*, pages 344–352. IEEE, 1991.

[10] Martin Otto. *Bisimulation Invariance and Finite Models*. Springer, 2006.

[11] Prakash Panangaden. *Labelled Markov Processes*. Imperial College Press, 2009.

[12] Dirk Pattinson. Coalgebraic modal logic: Soundness, completeness and decidability. *Theoretical Computer Science*, 2003.

[13] Jan J. M. M. Rutten. Universal coalgebra: A theory of systems. *Theoretical Computer Science*, 249(1):3–80, 2000.

[14] Roberto Segala. *Modeling and Verification of Randomized Distributed Real-Time Systems*. PhD thesis, MIT, 1995.

[15] Johan van Benthem. Modal logic and classical logic. *Bibliopolis*, 1983.

[16] Franck van Breugel and James Worrell. A behavioural pseudometric for probabilistic transition systems. *Theoretical Computer Science*, 331(1):115–142, 2005.

[17] Cédric Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer, 2009.